



ANDERSON JUNIOR COLLEGE
2013 JC 2 PRELIMINARY EXAMINATIONS

NAME: _____ **PDG:** _____/12 **Register No:** _____

CHEMISTRY

9647/02

Higher 2

16 September 2013

Paper 2 Structured Questions

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use				
Paper 2	1		Total	/ 72
	2			
	3			
	4			
	5			
	6			

This document consists of **20** printed pages and **1** blank page.

1 Planning (P)

The enthalpy change of neutralisation of a mixture of acid and base can be determined by measuring the temperature change when the two reactants are mixed together.

A student was asked to investigate the enthalpy change of neutralisation between malonic acid, $\text{HO}_2\text{CCH}_2\text{CO}_2\text{H}$ and aqueous sodium hydroxide. She was given two separate solutions, **FA 1** and a standard solution of NaOH.

FA 1 contains 1.00 mol dm^{-3} malonic acid

To ensure that both acidic protons in the malonic acid were reacted, she mixed 40 cm^3 of the 0.20 mol dm^{-3} solution of NaOH provided with 20 cm^3 of diluted **FA 1** and recorded the temperature change. The temperature change recorded was approximately 2°C and the student decided that it was too low and would create significant uncertainty in the results.

To overcome the problem, she decided to use the undiluted **FA 1** in the next measurement and prepared a more concentrated solution of NaOH, **FA 2** from solid anhydrous NaOH.

During the preparation, the student found that the process of dissolving NaOH released heat and the resultant **FA 2** solution had a significantly higher initial temperature than that of the malonic acid solution. When conducting the experiment, a weighted average of the initial temperatures of the two solutions was found to ensure that the initial temperature of the acid–base mixture is more accurate.

- (a) Construct a balanced equation, including the state symbols, for the reaction between malonic acid and aqueous sodium hydroxide.

..... [1]

- (b) The enthalpy change of neutralisation between malonic acid and aqueous sodium hydroxide may be determined by performing an experiment to measure the temperature change and then analysing the results.

Write a plan on the following page for such an experiment. In your plan, you should use the same proportions of malonic acid and NaOH solutions as described above.

You may assume that you are provided with the following.

- **FA 1**, having the concentration given above
- 12 g solid anhydrous sodium hydroxide
- deionised water
- the apparatus normally found in a school or college laboratory

Your plan should include the following.

- the intended concentration of **FA 2** to be used
- details for the preparation of this **FA 2** solution from the solid anhydrous sodium hydroxide provided
- a diagram of the assembled apparatus that you would use when measuring the temperature change
- all essential experimental details

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[6]

- (c) (i) A student carried out the experiment you have planned in (b) and the following data was collected.

Initial temperature of FA 1 / °C	a
Initial temperature of FA 2 / °C	b
Maximum temperature reached by mixture / °C	c

Outline how you would use her results to determine the enthalpy change of neutralisation between malonic acid and aqueous sodium hydroxide.

[You may assume that the heat capacity per unit volume of the mixture is $4.2 \text{ J K}^{-1} \text{ cm}^{-3}$]

- (ii) Using your answer to (c)(i) and given that the enthalpy change of solution of solid anhydrous NaOH is $-44.51 \text{ kJ mol}^{-1}$, sketch an energy cycle involving the reaction of solid anhydrous NaOH with malonic acid and determine its enthalpy change.

[4]

- (d) Identify one potential safety hazard in this experiment and state how you would minimise this risk.

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[1]

[Total: 12]

2 This question involves chlorine and its compounds.

(a) When chlorine is bubbled into aqueous potassium iodide, a brown mixture is formed.

(i) Write a balanced equation for the above reaction.

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(ii) Upon shaking the mixture with an equal volume of trichloroethane, two immiscible layers are observed.

Suggest, with reason, the colour of the organic layer.

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[3]

(b) Chlorine dissolves in the water to form chloric(I) acid (HOCl), the active agent for swimming pool disinfection, and hydrochloric acid.

A slightly alkaline pH of about 7.5 is desirable in swimming pools because this range is most comfortable to the human eye while maintaining water that is not corrosive.

The pH of the water in the swimming pools can be regulated by addition of HCl to control the amount of ClO^- and HOCl present in the solution. It is also recommended that the amount of HOCl present in the pool should not exceed 2 ppm.

(i) Given that $1 \text{ ppm} = 1 \text{ mg}$ of solute dissolved in 1000 g of solvent, show that the recommended concentration of HOCl present in the pool is $3.8 \times 10^{-5} \text{ mol dm}^{-3}$.
You may assume that the density of the solvent is 1 g cm^{-3} .

(ii) Write an expression for the acid dissociation constant of HOCl .

.....

- (iii) Calculate the concentration of ClO^- , in mol dm^{-3} , needed to maintain the desirable pH in the pool.
[K_a of HOCl is $3.5 \times 10^{-8} \text{ mol dm}^{-3}$]

concentration of $\text{ClO}^- = \dots\dots\dots \text{mol dm}^{-3}$ [3]

- (c) Chlorine dioxide, ClO_2 , is produced industrially by reaction of NaClO_3 with HCl .

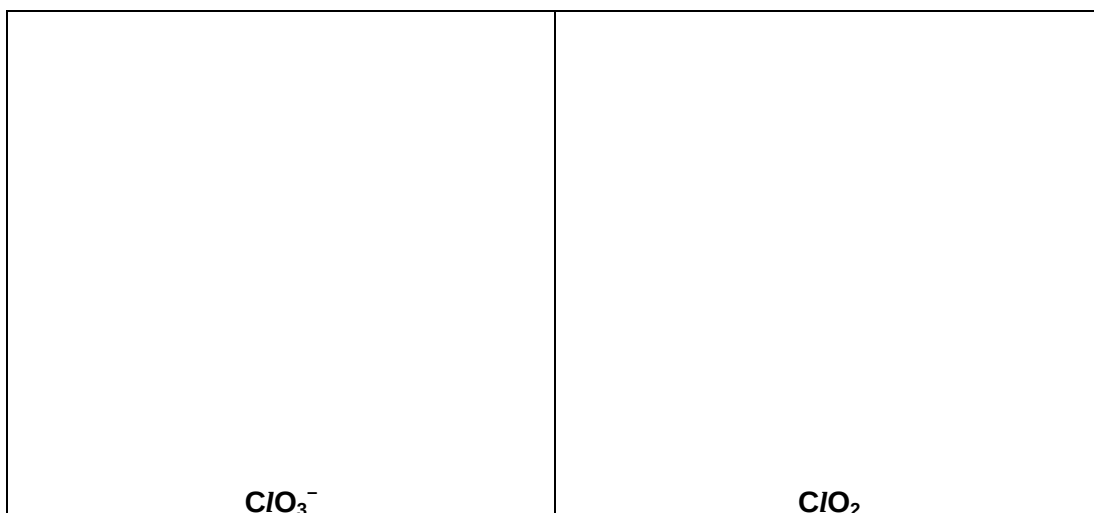
- (i) State the oxidation number (O.N.) of chlorine in NaClO_3 and ClO_2 .

O.N. of chlorine in NaClO_3

O.N. of chlorine in ClO_2

- (ii) By considering the numbers of bonding and non-bonding electron pairs, draw diagrams in the boxes below to show the likely shape of ClO_3^- and ClO_2 .

In your diagrams, state the shape and clearly indicate the value of bond angles for each.



[6]

(d) Chlorine dioxide is an unstable gas which decomposes into chlorine gas and oxygen gas.

(i) Write a balanced equation for this decomposition.

.....

(ii) What is the mass of chlorine gas formed when 150.0 g of chlorine dioxide is decomposed?

mass of chlorine gas formed = [2]
[Total: 14]

- 3** The Intel® super-fast 45 nm Core 2 processors are based on Penryn technology. This involves the use, for the first time in computer chips, of an oxide of hafnium.
- (a)** This oxide of hafnium has the formula HfO_2 . Calculate the percentage of hafnium by mass in this oxide.

percentage of hafnium by mass = [1]

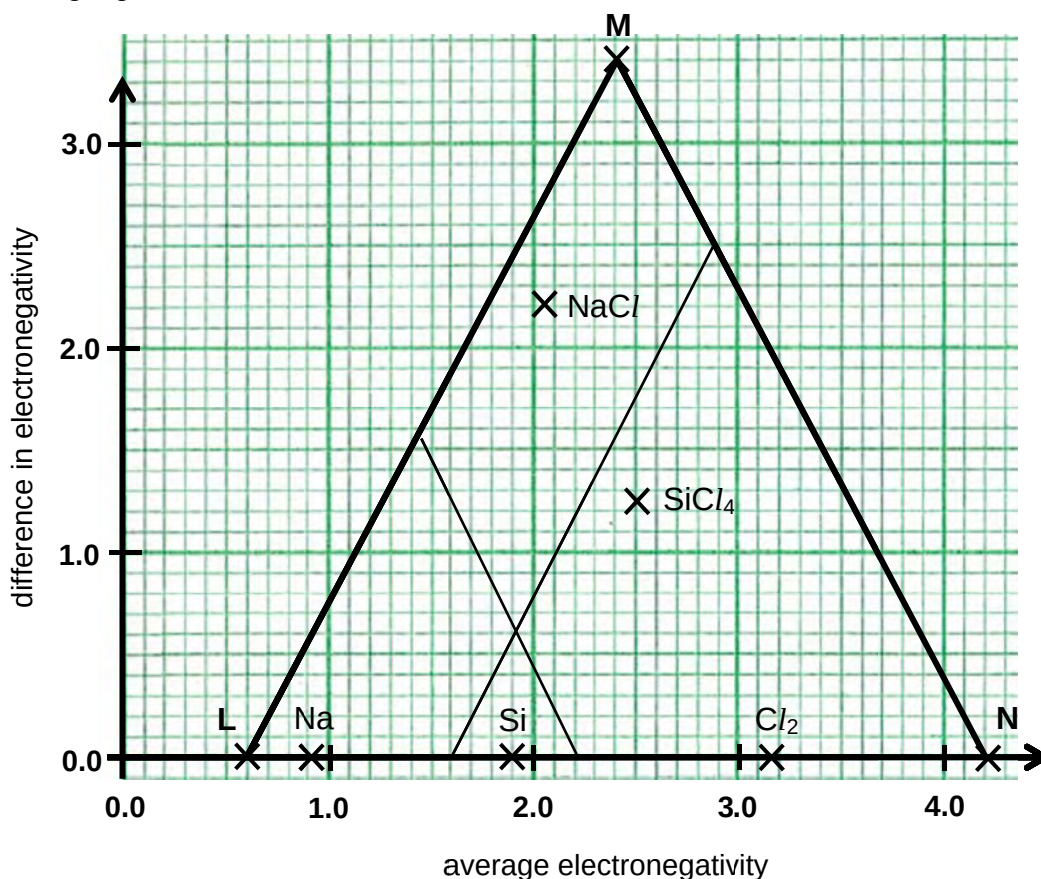
- (b) The type of bonding (covalent, ionic or metallic) present in a substance can be determined and predicted using the van Arkel triangle. The triangle is based on electronegativity values.

Difference in electronegativity between the element(s) is plotted along the y -axis and their average electronegativity is plotted along the x -axis.

The electronegativity data for O and Hf and selected elements from Period 3 of the Periodic Table are provided in the table below.

element	electronegativity
O	3.61
Na	0.93
Si	1.90
Cl	3.16
Hf	1.16

Some of the Period 3 elements and their compounds have been plotted on the van Arkel triangle given below.



- (i) Using your knowledge of the Period 3 elements and their compounds plotted above, state the type of bonding present at each of these bonding extremes, labelled **L**, **M** and **N** on the triangle?

L

M

N

- (ii) On the van Arkel triangle given on page 10, mark the point corresponding to the oxide of hafnium and use this point and your answer to **(b)(i)** to decide whether the oxide of hafnium is best described as ionic, covalent or metallic. Circle the correct option.

The oxide of hafnium is best described as

ionic covalent metallic

[4]

- (c) Computer chips contain electrical conductors, semiconductors and insulators.

Use your van Arkel plot in **(b)(ii)** to deduce the electrical properties of hafnium oxide. Circle the correct option.

The oxide of hafnium is a / an

conductor semiconductor insulator

[1]

[Total: 6]

- 4 A sulfur-containing compound which is present in the Earth's atmosphere is carbonyl sulfide, OCS. The sequence of atoms in the molecule is oxygen-carbon-sulfur and the molecule is **not** cyclic.

- (a) (i) Draw a dot-and-cross diagram to show the bonding in the OCS molecule. Show outer electrons only.

- (ii) Suggest a value for the O-C-S bond angle.

..... [2]

The variation of the volume with temperature of a fixed mass of an ideal gas at constant pressure may be represented by a relationship known as Charles's law,

$$V = k T$$

where V is the volume of a gas, T is the temperature in Kelvin and k is a constant.

An experiment was carried out to attempt to verify this law using a gas, having the same shape as carbonyl sulfide.

- A specially adapted gas syringe was filled with a sample of this gas. The syringe was placed in a temperature controlled chamber at 25 °C and left for 5 minutes. The initial volume of gas at this temperature was **26.0 cm³**.
 - The temperature was adjusted and, after leaving for 5 minutes, the change in gas volume for the new temperature was recorded.
 - The experiment was repeated several times at different temperatures and the results recorded on the following page. Each change in gas volume at the different temperatures is determined with respect to the initial volume of gas at 25 °C.
- (b) Process the results in the table on page 13 to calculate the volume of the gas and the corresponding temperature in Kelvin to enable you to plot a graph to show their inter-relationship.

(Note 0 °C is 273 K).

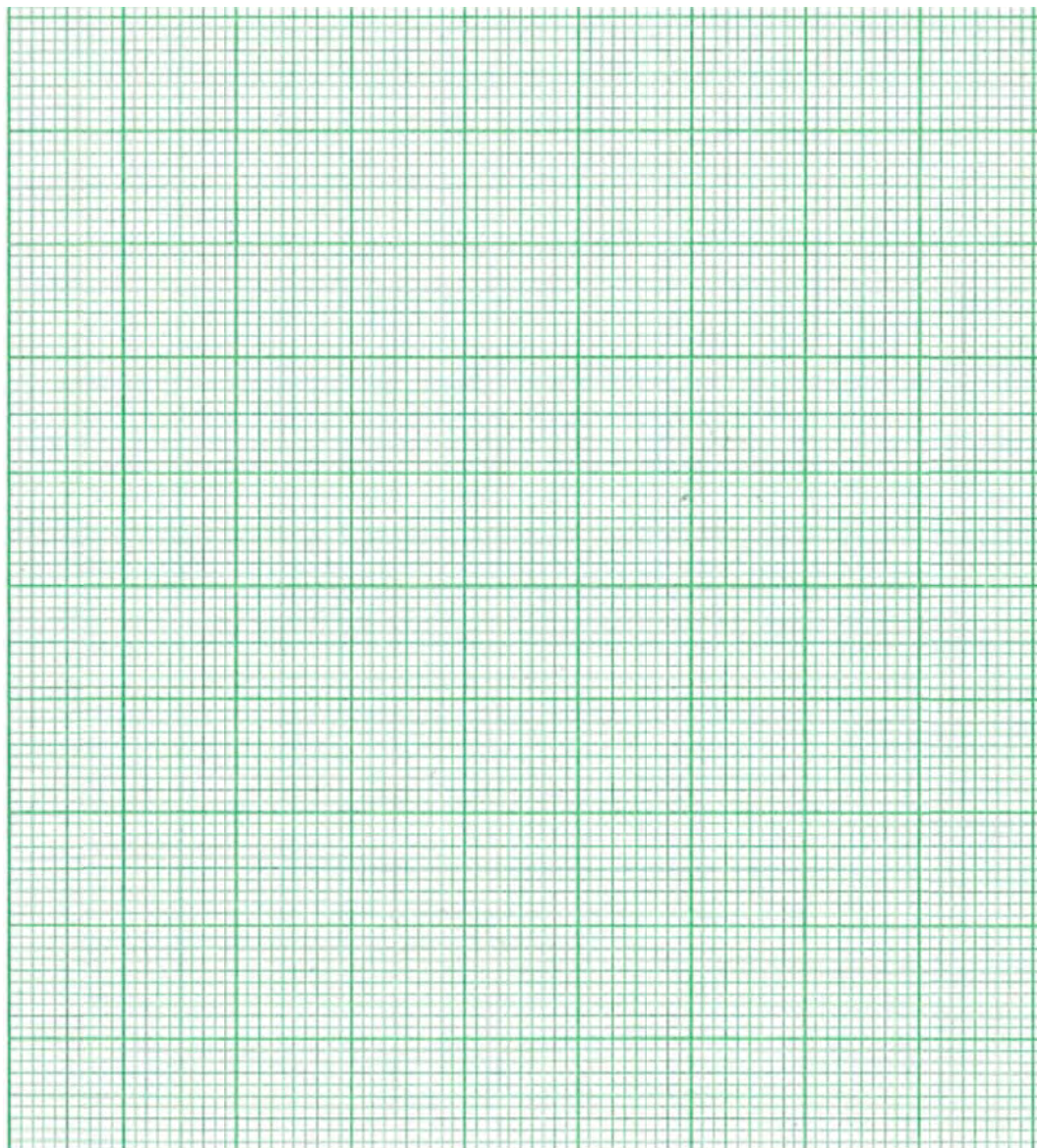
Record these values to **three significant figures** in the additional columns of the table.

Label the columns you use. For each column you use include units where appropriate.

temperature of the gas, T / $^{\circ}\text{C}$	change in volume of the gas, ΔV / cm^3		
-23	-4.20		
11	-2.10		
29	0.40		
63	3.30		
94	7.00		
131	9.20		

[2]

- (c) (i) Present the data calculated in (b) in graphical form. Draw the line of best fit.



- (ii) Using the graph obtained in (c)(i), indicate whether the results do or do not confirm Charles's law. Give a reason for your answer.

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[4]
[Total: 8]

- 5 (a) Use of the Data Booklet is relevant to parts of this question.

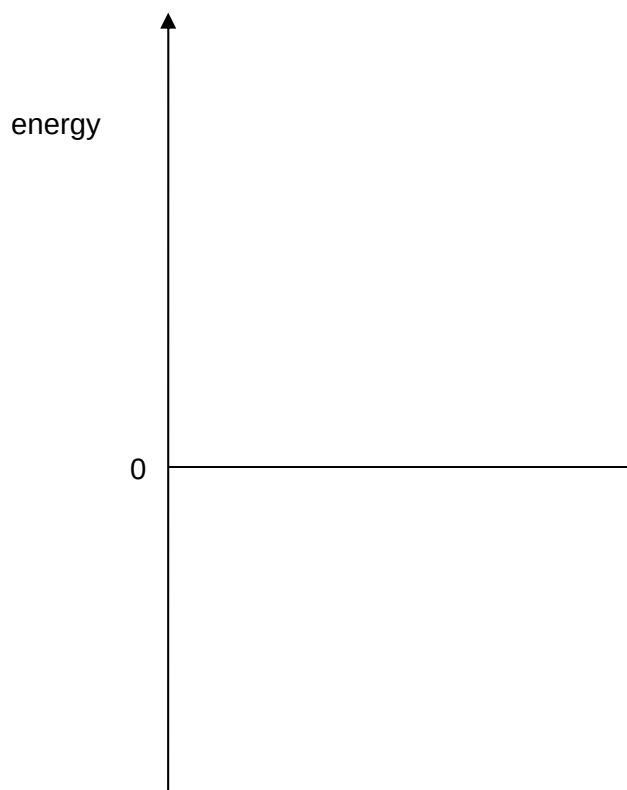
Calcium fluoride, CaF_2 , occurs naturally as the mineral fluorite and is a major source of fluorine.

- (i) Write an equation to represent the lattice energy of calcium fluoride.

.....

- (ii) Use the grid provided below and the following enthalpy changes to construct a Born–Haber cycle to calculate the lattice energy of calcium fluoride.

enthalpy change of formation of calcium fluoride	$-1220 \text{ kJ mol}^{-1}$
enthalpy change of atomisation of calcium	$+178 \text{ kJ mol}^{-1}$
enthalpy change of atomisation of fluorine	$+79 \text{ kJ mol}^{-1}$
first electron affinity of fluorine	-328 kJ mol^{-1}



lattice energy of calcium fluoride =

- (iii) How would you expect the lattice energy of calcium fluoride to compare with that of iron(II) fluoride? Explain your answer.

.....

[5]

- (b) Calcium compounds such as calcium fluoride and calcium hydroxide are sparingly soluble in water. Their solubility products at 298 K are given in the table below.

calcium compound	numerical value of K_{sp}
CaF_2	3.2×10^{-11}
Ca(OH)_2	6.4×10^{-6}

- (i) Calculate the solubility of calcium fluoride in water at 298 K.

solubility of calcium fluoride in water =

- (ii) A saturated solution **Q** containing calcium fluoride and calcium hydroxide has a pH of 12.

Calculate the solubility of calcium fluoride in solution **Q**.

solubility of calcium fluoride in solution **Q** =

- (iii) Comment on and explain the discrepancy in the solubilities of calcium fluoride in water and in solution Q.

.....

[5]

- (c) When a precipitate is formed, $\Delta G_{\text{ppt}}^{\ominus}$, in J mol^{-1} , is given by the following expression.

$$\Delta G_{\text{ppt}}^{\ominus} = 2.303RT \log_{10} K_{\text{sp}}$$

- (i) Use the data above to calculate $\Delta G_{\text{ppt}}^{\ominus}$, in kJ mol^{-1} , for calcium fluoride.

$$\Delta G_{\text{ppt}}^{\ominus} = \dots\dots\dots \text{kJ mol}^{-1}$$

- (ii) The standard enthalpy change for the formation of the precipitate $\text{CaF}_2(\text{s})$ is -224 kJ mol^{-1} .
 Use your answer in (c)(i) to calculate the $\Delta S_{\text{ppt}}^{\ominus}$, in $\text{J mol}^{-1} \text{ K}^{-1}$, for the formation of the precipitate $\text{CaF}_2(\text{s})$ at 298 K.

$$\Delta S_{\text{ppt}}^{\ominus} = \dots\dots\dots \text{J mol}^{-1} \text{ K}^{-1}$$

- (iii) Explain the significance of the sign of your answer in (c)(ii).

.....

[4]

- (d) Another common fluoride compound, silver fluoride, has a K_{sp} value of $1.006 \text{ mol}^2 \text{ dm}^{-6}$ at 298 K.

Use the expression given in (c) to justify whether silver fluoride is soluble in water at 298 K.

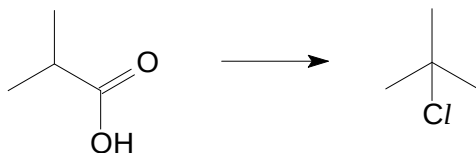
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[Total: 16]

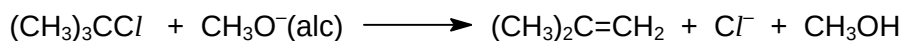
6 This question examines the chemistry of 2-chloro-2methylpropane, otherwise known as *t*-butyl chloride.

- (a) In **no more than 3 steps**, suggest how *t*-butyl chloride can be synthesised from 2-methylpropanoic acid. You are required to state the reagents and conditions for each step and give the structures of the intermediates formed.



[5]

t-butyl chloride, undergoes an elimination reaction when heated under reflux at 80 °C with alcoholic sodium methoxide to yield 2-methylpropene according to the following equation.



It was proposed that the reaction takes place via a two-step mechanism and is expected to obey the following rate law.

$$\text{rate} = k [\text{RX}]$$

where RX denotes an alkyl halide

The kinetics of this reaction was studied and the following results were obtained.

experiment	$[(\text{CH}_3)_3\text{CCl}] / \text{mol dm}^{-3}$	$[\text{CH}_3\text{O}^-] / \text{mol dm}^{-3}$	relative initial rate
1	0.060	0.060	1.00
2	0.050	0.060	0.833
3	0.040	0.050	0.667

- (b) (i) Use the data given above to deduce the rate law and explain if the above proposed rate law is correct.

(ii) Hence, propose a mechanism for the above elimination reaction given that the reaction proceeds via the following 2 steps:

- ☺ rate-determining step yields a carbocation;
- ☺ followed by acid-base reaction between H in the carbocation and CH_3O^- .

[4]

(c) Explain the effect on the rate constant for this reaction if

(i) the reaction mixture is now heated in an oil bath at 100°C .

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(ii) *t*-butyl bromide, $(\text{CH}_3)_3\text{CBr}$, is used instead.

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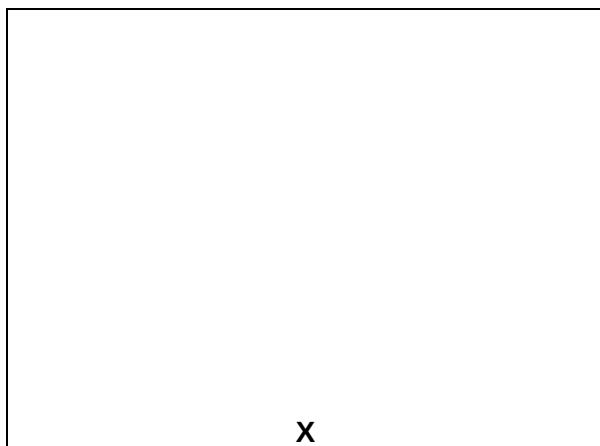
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[4]

- (d) When *t*-butyl chloride reacts with hot alcoholic sodium methoxide, a by-product, **X** ($M_r = 88.0$), is formed.

Suggest the structure of **X**.



[1]

- (e) *t*-butyl amine may be synthesised from the reaction of *t*-butyl chloride with excess concentrated NH_3 .

- (i) Suggest why there is a need to use excess concentrated NH_3 .

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- (ii) In the synthesis of *t*-butyl amine, an impurity with molecular formula $\text{C}_8\text{H}_{19}\text{N}$ was formed. Suggest a structure for this impurity.

[2]

[Total: 16]